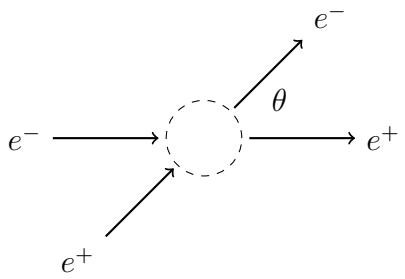
Bhabha scattering 1

Bhabha scattering is the process $e^- + e^+ \to e^- + e^+$. The arrows for e^+ are reversed because a positron is treated mathematically as an electron moving backwards in time.



Momentum vectors where $E = \sqrt{p^2 + m^2}$.

$$p_1 = \underset{e^- \longrightarrow}{\begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix}} \qquad p_2 = \underset{\longrightarrow e^+}{\begin{pmatrix} E \\ 0 \\ 0 \\ -p \end{pmatrix}} \qquad p_3 = \underset{\nearrow}{\overset{e^-}{\begin{pmatrix} E \\ p\sin\theta\cos\phi \\ p\sin\theta\sin\phi \\ p\cos\theta \end{pmatrix}}} \qquad p_4 = \underset{e^+ \nearrow}{\begin{pmatrix} E \\ -p\sin\theta\cos\phi \\ -p\sin\theta\sin\phi \\ -p\cos\theta \end{pmatrix}}$$

Spinors for p_1.

$$u_{11} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ -p \\ 0 \end{pmatrix}} \qquad u_{12} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ 0 \\ p \end{pmatrix}}$$

Spinors for p_2.

$$v_{21} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} p \\ 0 \\ E+m \\ 0 \end{pmatrix}} \qquad v_{22} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ -p \\ 0 \\ E+m \end{pmatrix}}$$

Spinors for p_3.

$$u_{31} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ p_{3z} \\ p_{3x} + ip_{3y} \end{pmatrix}} \qquad u_{32} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ p_{3x} - ip_{3y} \\ -p_{3z} \end{pmatrix}}$$

Spinors for p_4.

$$v_{41} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} p_{4z} \\ p_{4x} + ip_{4y} \\ E+m \\ 0 \end{pmatrix}} \qquad v_{42} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} p_{4x} - ip_{4y} \\ -p_{4z} \\ 0 \\ E+m \end{pmatrix}}$$

The scattering amplitude $\mathcal{M}$ is

$$\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$$

where $\mathcal{M}_1$ is the $1 \to 3$, $4 \to 2$ amplitude

$$\mathcal{M}_1 = \frac{e^2}{t}(\bar{v}_{2b}\gamma^\mu v_{4d})(\bar{u}_{3c}\gamma_\mu u_{1a})$$

and $\mathcal{M}_2$ is the $1 \to 2$, $4 \to 3$ amplitude

$$\mathcal{M}_2 = \frac{e^2}{s}(\bar{u}_{3c}\gamma^\mu v_{4d})(\bar{v}_{2b}\gamma_\mu u_{1a})$$

The expected density $\langle|\mathcal{M}|^2\rangle$ is the average over spin states.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{abcd}|\mathcal{M}|^2 = \frac{1}{4}\sum_{abcd}(\mathcal{M}_1 - \mathcal{M}_2)(\mathcal{M}_1 - \mathcal{M}_2)^*$$

Expand the summand.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{abcd}(\mathcal{M}_1\mathcal{M}_1^* - \mathcal{M}_1\mathcal{M}_2^* - \mathcal{M}_2\mathcal{M}_1^* + \mathcal{M}_2\mathcal{M}_2^*)$$

The summation results are

$$\sum_{abcd}\mathcal{M}_1\mathcal{M}_1^* = \frac{e^4}{t^2}\left(8s^2 + 8u^2 - 64m^2s - 64m^2u + 192m^4\right)$$
$$\sum_{abcd}\mathcal{M}_1\mathcal{M}_2^* = \frac{e^4}{st}\left(-8u^2 + 64m^2u - 96m^4\right)$$
$$\sum_{abcd}\mathcal{M}_2\mathcal{M}_2^* = \frac{e^4}{s^2}\left(8t^2 + 8u^2 - 64m^2t - 64m^2u + 192m^4\right)$$

where s, t, and u are Mandelstam variables

$$s = (p_1 + p_2)^2$$
$$t = (p_1 - p_3)^2$$
$$u = (p_1 - p_4)^2$$

For clarity define

$$f_{11} = 8s^2 + 8u^2 - 64m^2s - 64m^2u + 192m^4$$
$$f_{12} = -8u^2 + 64m^2u - 96m^4$$
$$f_{22} = 8t^2 + 8u^2 - 64m^2t - 64m^2u + 192m^4$$

and write $\langle|\mathcal{M}|^2\rangle$ more compactly as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(\frac{f_{11}}{t^2} - \frac{2f_{12}}{st} + \frac{f_{22}}{s^2}\right)$$

For $E \gg m$ we use the approximation $m = 0$ and obtain

$$\begin{aligned} f_{11} &= 8s^2 + 8u^2 \\ f_{12} &= -8u^2 \\ f_{22} &= 8t^2 + 8u^2 \end{aligned}$$

Then $\langle|\mathcal{M}|^2\rangle$ simplifies as

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{s^2 + u^2}{t^2} + \frac{2u^2}{st} + \frac{t^2 + u^2}{s^2} \right)$$

For $m = 0$ the Mandelstam variables are

$$\begin{aligned} s &= 4E^2 \\ t &= -2E^2(1 - \cos\theta) \\ u &= -2E^2(1 + \cos\theta) \end{aligned}$$

Hence

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{1 + \cos^4(\theta/2)}{\sin^4(\theta/2)} - \frac{2\cos^4(\theta/2)}{\sin^2(\theta/2)} + \frac{1 + \cos^2\theta}{2} \right)$$

This can be written more compactly as

$$\langle|\mathcal{M}|^2\rangle = e^4 \left(\frac{\cos^2\theta + 3}{\cos\theta - 1} \right)^2$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{e^4} \frac{\langle|\mathcal{M}|^2\rangle}{4s} = \frac{\alpha^2}{16E^2} \left(\frac{\cos^2\theta + 3}{\cos\theta - 1} \right)^2$$

Multiply by $(\hbar c)^2$ to convert from natural units to physical units.

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{16E^2} \left(\frac{\cos^2\theta + 3}{\cos\theta - 1} \right)^2 (\hbar c)^2$$

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